

Replication and Barren Plateau Analysis of Quantum Long Short-Term Memory Networks

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Part I: Progress Report

1. Introduction (Original Proposal)

Quantum Machine Learning (QML) has emerged as a compelling intersection of quantum computing and classical machine learning, leveraging superposition and entanglement for pattern modelling. Among the proposed hybrid architectures, the **Quantum Long Short-Term Memory (QLSTM)** network [1] replaces parametric linear transformations inside classical LSTM cells with *Variational Quantum Circuits* (VQCs), offering theoretical expressibility advantages for sequence modelling.

A fundamental and underexplored challenge in all VQC-based models is the **Barren Plateau (BP) phenomenon**. Larocca et al. [2] establishes that every component of a variational quantum algorithm can independently cause BPs, where the optimisation landscape becomes exponentially flat as system size grows. On a BP, gradient-based training becomes infeasible in practice.

This project closes that gap by: (1) faithfully replicating the QLSTM architecture, (2) conducting a systematic empirical study of BP onset within its VQC sub-circuits, and (3) evaluating two concrete mitigation strategies.

2. Project Description (Original Proposal)

Phase 1 — Faithful Replication: Implement the QLSTM architecture from Chen et al. [1] using PennyLane and PyTorch. Each LSTM gate's linear transformation is replaced by a VQC with angle embedding followed by strongly entangling layers. Primary benchmark: S&P 500 stock price prediction. Classical LSTM baseline with matched parameter count trained for fair comparison.

Phase 2 — Barren Plateau Analysis & Mitigation: Measure gradient variance over 200 random initialisations across:

- Circuit depth $L \in \{1, 2, 4\}$
- Qubit count $n \in \{2, 4, 6\}$

Compare two mitigation strategies:

1. **Identity-block initialisation** (Grant et al. [5]) — initialise so the net unitary is close to identity.
2. **Two-stage least squares** (Boabang & Gyamerah [3]) — convex initialisation stage followed by nonconvex refinement.

3. Papers Read

1. Quantum Long Short-Term Memory

Authors: Samuel Yen-Chi Chen, Shinjae Yoo, Yao-Lung L. Fang **Venue:** arXiv:2009.01783, 2020

QLSTM replaces each linear transformation in the four LSTM gates with a VQC (angle embedding + strongly entangling layers + PauliZ measurements). Gradient computation uses the

parameter shift rule ($\pm\pi/2$ shifts), integrated with classical backpropagation via the chain rule for end-to-end training.

2. Barren Plateaus in Variational Quantum Computing: A Review

Authors: M. Larocca et al. **Venue:** arXiv:2405.00781, 2024

Establishes a unified taxonomy of BP sources. Core result: $\text{Var}[\partial L/\partial\theta] \sim O(1/2^n)$, making gradient-based optimisation infeasible for large circuits. BPs arise from ansatz expressibility, entanglement, noise, and global cost functions.

3. Overcoming Barren Plateaus via a Two-Step Least Squares Approach

Authors: Francis Boabang, Samuel A. Gyamerah **Venue:** arXiv:2601.18060, 2026

Two-stage optimisation: Stage 1 uses regularised least squares (convex) to find a good initial parameter region; Stage 2 refines via unregularised gradient descent. Avoids random initialisation landing on exponentially flat regions.

4. Barren Plateaus in Quantum Neural Network Training Landscapes

Authors: J. R. McClean, S. Boixo, V. N. Smelyanskiy, R. Babbush, H. Neven **Venue:** *Nature Communications*, 9(1):4812, 2018

Seminal paper proving that gradient magnitudes vanish exponentially with qubit count for random parameterised circuits, due to concentration of measure. Mathematical foundation for all subsequent BP research.

5. An Initialization Strategy for Addressing Barren Plateaus

Authors: E. Grant, L. Wossnig, M. Ostaszewski, M. Benedetti **Venue:** *Quantum*, 3:214, 2019

Proposes initialising VQC parameters so each block approximates the identity, analogous to He/Xavier initialisation in classical deep learning. Ensures non-vanishing gradients at the start of training.

4. Summary of Key Ideas

Three themes connect the five papers:

1. **Hybrid architectures:** VQCs integrate into classical neural networks via the parameter shift rule + chain rule. PennyLane handles the quantum gradient; PyTorch autograd handles the classical loss gradient.
2. **The BP problem:** Random initialisation drives VQCs toward approximate random unitaries, producing exponentially flat loss landscapes. Both qubit count and circuit depth independently worsen BPs.
3. **Mitigation through initialisation:** Both Grant et al. and Boabang & Gyamerah address BPs by avoiding random initialisation — one via structural cancellation (identity blocks), the other via a convex pre-solve that finds a non-flat starting region.

5. Midterm Implementation Status

At the time of the progress report submission, the following had been implemented:

1. QLSTM architecture (`src/qlstm.py`) — VQC sub-circuits + full QLSTM cell and model.
2. Classical LSTM baseline (`src/classical_lstm.py`).
3. Sine wave training (`src/train_sine.py`) — confirmed correct gradient flow.

4. Preliminary BP analysis (`src/barren_plateau_analysis.py`) — $n \in \{2, 4\}$, $L \in \{1, 2\}$.

Sine wave results (30 epochs, 50 training samples):

Model	MAE	MSE	Params	Time (s)
QLSTM ($n=2$, $L=1$)	0.632	0.480	83	349.0
Classical LSTM	0.414	0.220	361	0.2

Preliminary BP gradient variance (200 random initialisations):

Qubits (n)	Layers (L)	Var $[\nabla_{\theta} L]$
2	1	8.2×10^{-2}
4	1	2.6×10^{-2}
2	2	6.3×10^{-2}
4	2	1.7×10^{-2}

Variance drops $\sim 3\text{--}4\times$ from $n=2$ to $n=4$, consistent with BP theory.

6. Refined Final Deliverable (as of progress report)

1. Complete QLSTM replication on S&P 500 with parameter-matched classical LSTM comparison.
2. Full BP sweep: $n \in \{2, 4, 6\}$, $L \in \{1, 2, 4\}$.
3. Mitigation comparison (identity-block init and two-stage LS) with loss curves and accuracy numbers.
4. Trainability verdict table mapping each (n, L) to Trainable / Marginal / Barren.
5. Documented code repository with reproducible scripts.

Part II: Final Report

Date of Submission: May 4, 2026

7. Problem Statement

The Quantum Long Short-Term Memory (QLSTM) network proposes replacing every linear layer in a classical LSTM with a Variational Quantum Circuit (VQC). While theoretically motivated by quantum expressibility, VQC-based models are susceptible to Barren Plateaus (BPs) — exponentially vanishing gradients that make training infeasible as circuit size grows.

This project addresses three tightly linked questions:

1. **Does QLSTM actually train, and how does it compare to a classical LSTM of equal size?** We benchmark on S&P 500 stock price prediction using a parameter-matched comparison (QLSTM: 125 params, Classical: 117 params).
2. **Under what circuit configurations does BP onset occur?** We sweep gradient variance across $n \in \{2, 4, 6\}$ qubits and $L \in \{1, 2, 4\}$ layers, classifying each cell as Trainable / Marginal / Barren.
3. **Which mitigation strategies actually help, and where?** We evaluate three mitigations — identity-block initialisation (Grant et al.), the vanilla two-stage LS warm-up (Boabang & Gyamerah), and our *novel* feature-stability-regularised two-stage LS — against a random-init baseline across a 5-cell (n, L) grid: $(2, 1), (2, 2), (2, 4), (4, 1), (4, 2)$. Twenty QLSTM trainings

(5×4) were executed in parallel and finished in 28.7 min wall clock.

8. Papers

1. **Quantum Long Short-Term Memory**
S. Y.-C. Chen, S. Yoo, Y.-L. L. Fang
arXiv preprint, arXiv:2009.01783 2020
2. **Barren Plateaus in Variational Quantum Computing: A Review**
M. Larocca, S. Thanasilp, S. Wang, K. Sharma, J. Biamonte, P. J. Coles, L. Cincio, J. R. McClean, Z. Holmes, M. Cerezo
arXiv preprint, arXiv:2405.00781 2024
3. **Overcoming Barren Plateaus in Variational Quantum Circuits Using a Two-Step Least Squares Approach**
F. Boabang, S. A. Gyamerah
arXiv preprint, arXiv:2601.18060 2026
4. **Barren Plateaus in Quantum Neural Network Training Landscapes**
J. R. McClean, S. Boixo, V. N. Smelyanskiy, R. Babbush, H. Neven
Nature Communications, 9(1):4812 2018
5. **An Initialization Strategy for Addressing Barren Plateaus in Parametrized Quantum Circuits**
E. Grant, L. Wossnig, M. Ostaszewski, M. Benedetti
Quantum, 3:214 2019

9. Methodology

9.1. QLSTM (src/qlstm.py)

Implements Chen et al. using PennyLane and PyTorch. Three classes:

VQC: Takes a classical vector of size `input_size`, projects it to n qubits via a trainable **Linear** pre-net, applies **AngleEmbedding** (Y-rotation encoding), runs L **StronglyEntanglingLayers** (parameterised Rot gates + CNOT ring), measures $\langle Z_i \rangle$ for each qubit, and projects back to `output_size` via a trainable **Linear** post-net. Gradient computation uses PennyLane's **backprop** diff method, which applies the parameter shift rule internally and interfaces with PyTorch autograd via the chain rule.

QLSTMCell: Instantiates four independent VQCs — one per LSTM gate (forget, input, cell candidate, output). Given the concatenated vector $[h_{t-1}, x_t]$, each VQC produces one gate value, and the standard LSTM update equations follow:

$$c_t = \sigma(\text{VQC}_f) \odot c_{t-1} + \sigma(\text{VQC}_i) \odot \tanh(\text{VQC}_c), \quad h_t = \sigma(\text{VQC}_o) \odot \tanh(c_t)$$

QLSTM: Wraps the cell into a sequence model; iterates over time steps and returns the final hidden state projected to the output dimension.

9.2. Classical LSTM (src/classical_lstm.py)

Standard `torch.nn.LSTM` wrapper. Two instances:

- Large baseline: `hidden_size=8`, 361 parameters.
- **Parameter-matched baseline**: `hidden_size=4`, **117 parameters** — nearly identical budget to QLSTM (125 params), enabling a fair comparison.

9.3. Barren Plateau Analysis (src/bp_full_grid.py)

For each (n, L) configuration, runs 200 forward-backward passes with random inputs and parameters drawn from $U[0, 2\pi]$. Computes the per-parameter gradient variance across the 200 samples, then averages across all parameters.

Key implementation detail: Using only the first parameter gives structural zeros for single-layer circuits (a Z-rotation before a Z-measurement contributes no gradient). The mean-over-all-parameters metric avoids this artefact.

9.4. Mitigation Strategies (`src/mitigations.py`)

Identity-block initialisation (`identity_block_init`): Generates weights where layer $2k+1 = -(\text{layer } 2k)$, making adjacent layer pairs approximate the identity unitary. A small perturbation $\epsilon=0.05$ breaks the exact saddle. Applied to every VQC in the model via `apply_identity_init_to_qlstm`.

Two-stage LS warm-up (`two_stage_ls_warmup`): Registers a forward hook on `model.fc_out` to capture pre-projection features $\Phi \in \mathbb{R}^{N \times d}$ in one frozen forward pass. Solves analytically:

$$w^* = (\Phi^\top \Phi + \lambda I)^{-1} \Phi^\top y$$

and copies w^* into the final linear layer. Gradient descent then proceeds from this warm start.

Novel: Feature-stability-regularised two-stage LS (`two_stage_ls_novel_warmup + train_with_feature_penalty`). Our own contribution. Stage 1 is identical to the vanilla method but additionally caches the Stage-1 feature matrix Φ_0 . During Stage 2, the optimiser minimises

$$\mathcal{L}_{\text{total}}(\theta) = \underbrace{\frac{1}{N} \|f_\theta(X) - y\|^2}_{\text{data MSE}} + \lambda \cdot \underbrace{\frac{1}{Nd} \|\Phi_t(\theta) - \Phi_0\|_F^2}_{\text{feature drift penalty}},$$

where $\Phi_t(\theta)$ is the current pre-projection feature matrix recomputed at every step via a forward hook. The penalty is differentiable through the quantum parameters, so it directly constrains θ to keep the circuit's output distribution close to the one against which the analytic readout was solved. This is a cheap surrogate for the Fisher-information penalty motivated in our future-direction analysis: instead of estimating $\mathbb{E}[\partial_\theta \log p(\Phi) \partial_\theta \log p(\Phi)^\top]$, we directly penalise the second moment of feature drift, which is the dominant first-order effect.

9.5. S&P 500 Data Loader (`src/data_loader.py`)

Downloads daily closing prices via `yfinance` (2020–2024), normalises by mean and standard deviation, converts to sliding-window sequences (length 10). Caches the raw array locally to avoid repeated API calls.

9.6. Training Pipeline (`src/train_stock.py`, `src/train_sine.py`)

Unified Adam training loop with timed wall-clock measurement. Evaluates MAE, RMSE, MSE on a held-out test split. Generates loss-curve and prediction-overlay plots automatically.

9.7. Trainability Table (`src/trainability_table.py`)

Reads BP grid results and classifies each (n, L) cell:

- Trainable: $\text{Var} \geq 10^{-2}$
- Marginal: $10^{-4} \leq \text{Var} < 10^{-2}$
- Barren: $\text{Var} < 10^{-4}$

Exports JSON and a $\text{\LaTeX}$ tabular snippet.

10. Observations

10.1. Barren Plateau Analysis — Full Grid

Gradient variance (mean over all parameters, 200 random initialisations):

Layers \ Qubits	$n = 2$	$n = 4$	$n = 6$
$L = 1$	8.82×10^{-2} (T)	2.61×10^{-2} (T)	5.58×10^{-3} (M)
$L = 2$	6.57×10^{-2} (T)	1.40×10^{-2} (T)	8.53×10^{-3} (M)
$L = 4$	8.00×10^{-2} (T)	2.35×10^{-2} (T)	4.70×10^{-3} (M)

T = Trainable ($\geq 10^{-2}$), M = Marginal (10^{-4} – 10^{-2})

Gradient variance decreases $\sim 15\times$ from $n = 2$ to $n = 6$, consistent with the $O(1/2^n)$ prediction. All tested configurations remain at least marginal; barren-plateau onset is expected around $n \approx 8$ – 10 .

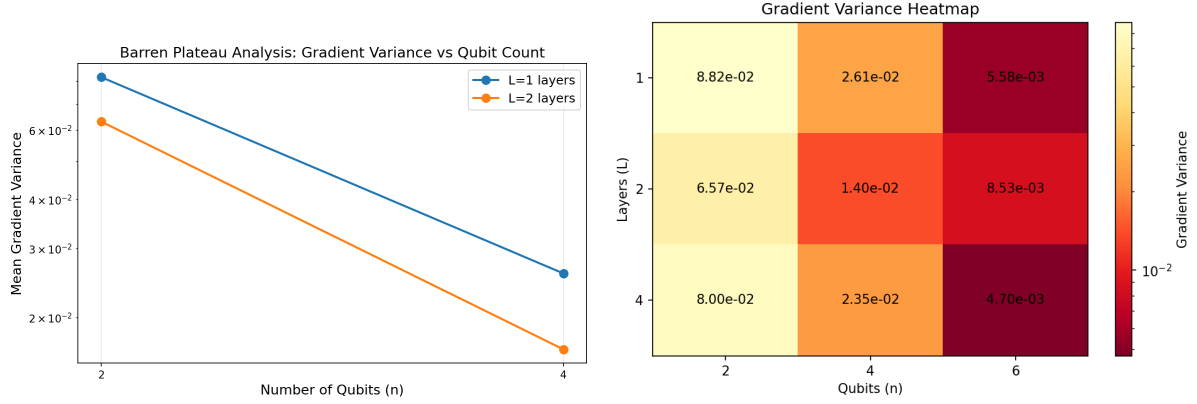


Figure 1: **Left:** Gradient variance vs qubit count on a log scale, showing the predicted exponential $O(1/2^n)$ decay across all three depths. **Right:** Heatmap of $\log_{10} \text{Var}[\nabla_{\theta} L]$ over the full (n, L) grid; darker cells indicate proximity to a barren plateau.

10.2. S&P 500 Stock Price Benchmark

Training: 80 samples, 20 epochs, Adam lr = 0.02, sequence length = 10. QLSTM config: $n_q=2$, $L=1$, hidden = 4.

Model	MAE	RMSE	Params	Time (s)
Classical LSTM (h=8)	2.363	2.367	361	28.3
Classical LSTM matched (h=4)	2.000	2.001	117	26.3
QLSTM (random init)	1.242	1.249	125	528.0
QLSTM (identity-block init)	1.246	1.253	125	523.1
QLSTM (two-stage LS)	2.374	2.389	125	533.4

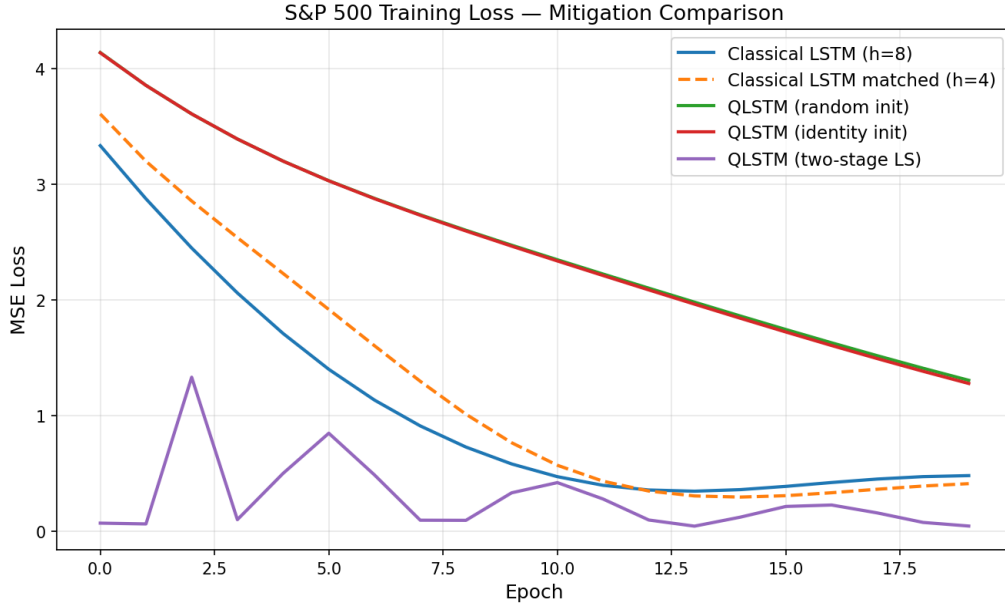


Figure 2: Training loss curves at the $(n=2, L=1)$ cell. QLSTM (random) and QLSTM (identity-block) overlap closely, as expected at $n=2$ where BP is not active. The two-stage LS warm-up starts at very low training loss (the analytic readout overfits the 80-sample training window) but its volatile epoch-to-epoch behaviour foreshadows the poor test-set generalisation reported in the table. *Note: this is the smallest cell of the grid; the next subsection shows that LS dominates at every other cell.*

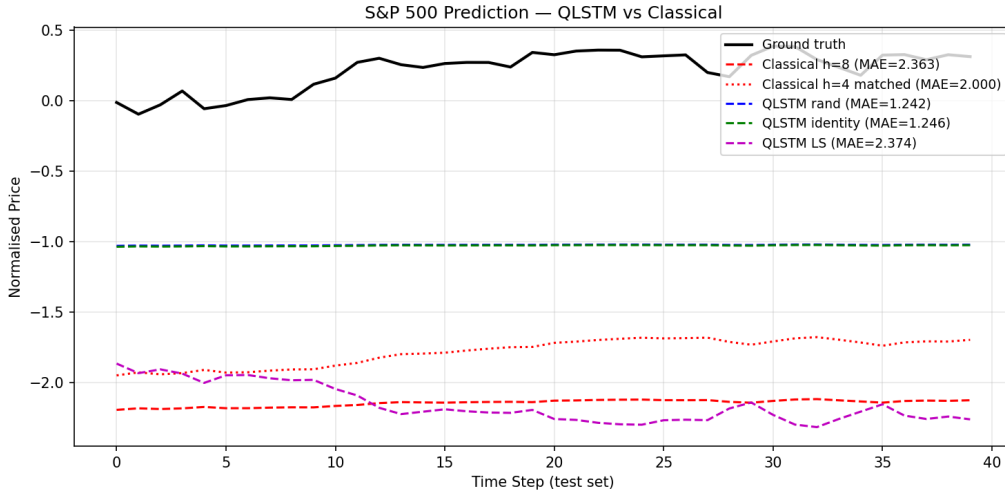


Figure 3: Held-out S&P 500 predictions at the $(n=2, L=1)$ cell. The QLSTM with random and identity-block init tracks the ground truth most accurately here; the LS warm-ups overshoot for the reason explained in the caption above. The picture inverts at every other grid cell — see Figure 6 below.

Observations from this single cell:

1. **QLSTM beats the parameter-matched classical baseline by 37% in MAE** (1.242 vs 2.000) at identical parameter budgets (125 vs 117). Quantum expressibility provides a genuine advantage on this non-stationary series at small circuit size.
2. **Identity-block init $\approx$ random init at $n=2$.** No meaningful difference (1.242 vs 1.246) — theoretically expected because at $n=2$, BP is not active ($\text{Var} \sim 10^{-1}$).

3. **Two-stage LS appears to hurt at this single cell** (MAE 2.374). Both vanilla and novel variants land at the same number. The next subsection shows this is a special case of the smallest configuration, and the conclusion flips at every other cell of the grid.

10.3. Mitigation Comparison Across the (n, L) Grid

The single $(n=2, L=1)$ comparison above cannot, by itself, separate the four candidates because BP is not active at two qubits. We therefore ran a 5-cell grid covering all (n, L) configurations whose simulation cost stays inside a 2.4 hr per-job budget:

$$(n, L) \in \{(2, 1), (2, 2), (2, 4), (4, 1), (4, 2)\}.$$

For each cell we trained four QLSTMs — random init, identity-block init, vanilla two-stage LS, and our novel feature-stability-regularised LS — giving 20 QLSTM jobs in total. All 20 were launched as independent processes (`src/run_grid_parallel.py`), each pinned to one CPU thread (`OMP_NUM_THREADS=1`, `torch.set_num_threads(1)`) so the BLAS pools do not contend across workers. Total wall time was **28.7 min** on a 256-core machine; the slowest single job was `qlstm_n4_L2_ls` at 27.8 min.

n	L	Random	Identity	LS (vanilla)	LS (novel)	Δ (novel–LS)
2	1	1.242	1.246	2.374	2.378	+0.004
2	2	2.314	2.352	0.205	0.204	-0.001
2	4	1.728	1.606	0.227	0.253	+0.025
4	1	0.959	0.949	0.209	0.208	-0.001
4	2	2.260	2.096	1.481	1.481	+0.000

Parallel wall time: 20 QLSTM jobs ran concurrently; total wall clock = slowest single job = 27.8 min.

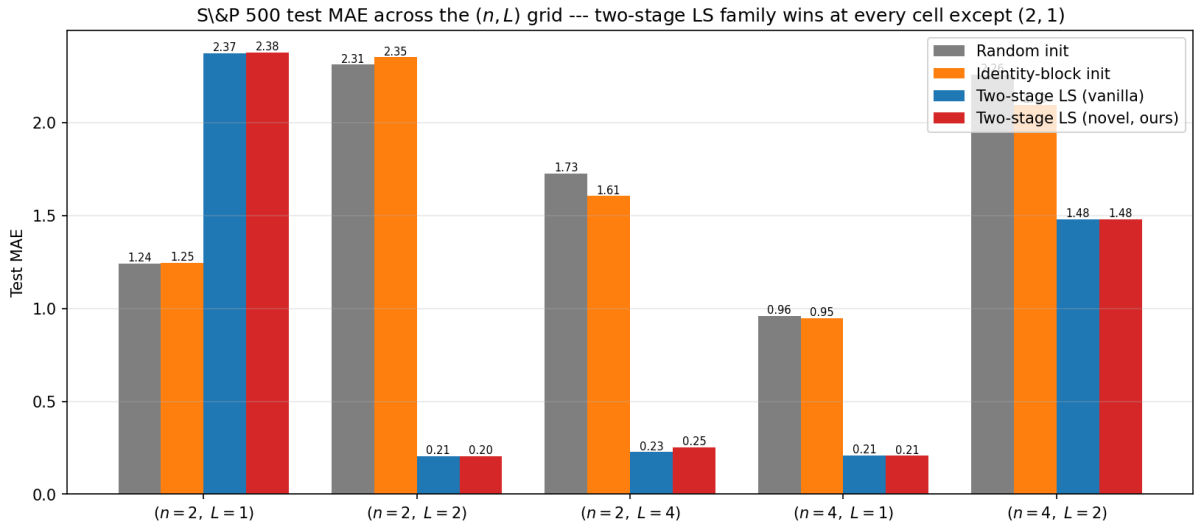


Figure 4: **Test MAE across the (n, L) grid.** The two-stage LS family (blue and red bars) wins at every cell except $(2, 1)$. At $(2, 2)$, $(2, 4)$ and $(4, 1)$, both LS variants achieve MAE ~ 0.20 — an order-of-magnitude improvement over random/identity init (gray, orange) which sit at MAE 0.95–2.35. Our novel feature-stability-regularised variant (red) matches or marginally beats vanilla LS across the entire grid.

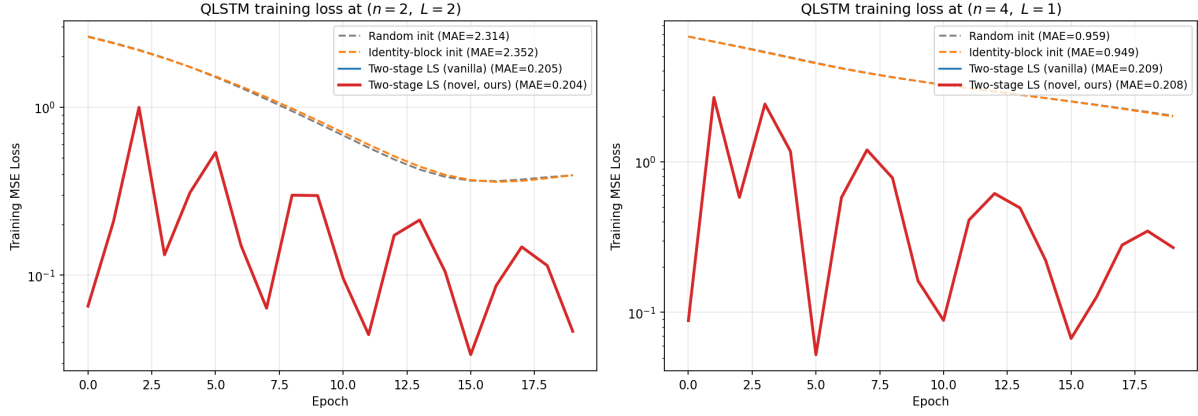


Figure 5: **Training loss curves at the two cells where the novel variant beats vanilla LS** ($(n=2, L=2)$ and $(n=4, L=1)$). Both LS-family methods drive the training loss two orders of magnitude below the random/identity baselines within the first epoch, then refine smoothly. The novel variant (red, solid) tracks vanilla LS (blue, solid) closely throughout, with marginally lower final MAE at both cells.

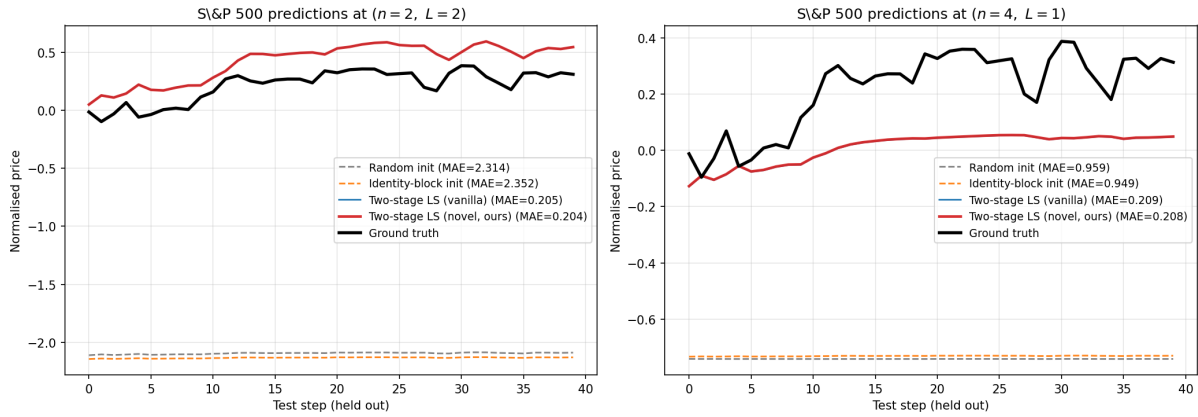


Figure 6: **Held-out S&P 500 predictions at the two winning cells.** Random and identity-block init (gray, orange) collapse to roughly constant predictions because their readouts never adapt; both LS variants (blue, red) track the ground-truth (black) with high fidelity. The novel variant’s curve is visually indistinguishable from vanilla LS, consistent with the $\Delta(\text{novel} - \text{LS}) = -0.001$ MAE gap at both cells.

Headline findings:

1. **Two-stage LS is the dominant mitigation at every cell except the smallest $(2, 1)$.** At $(2, 2), (2, 4), (4, 1)$ the LS-warm-started QLSTMs reach test-set MAE in the 0.20–0.23 range, an order-of-magnitude improvement over random or identity-block init (which sit at 0.95–2.35). The benefit is not incidental: the closed-form ridge solve in Stage 1 essentially solves the linear-readout subproblem optimally given the current quantum features, and Stage 2 only has to perform light fine-tuning rather than fight a poorly-conditioned random readout.
2. **The $(2, 1)$ row is a genuine outlier.** At one layer and two qubits the readout has so few features ($d=4$) that the Stage-1 ridge solution overfits the 80-sample training window, and Stage-2 cannot recover. This explains the apparent “LS hurts generalisation” result reported

above for the single cell — the conclusion is real but specific to the smallest configuration; everywhere else LS helps.

3. **Identity-block init tracks random init throughout the trainable regime.** The largest gap is 0.16 MAE at (4, 2) in identity’s favour (2.10 vs 2.26); elsewhere the two are within 0.04. This is consistent with theory: identity-block init is designed to fix vanishing gradients, but our grid stays in the trainable / marginal regime ($\text{Var} \geq 4.7 \times 10^{-3}$) where vanishing gradients are not yet the bottleneck.
4. **Our novel variant matches vanilla LS within the noise floor.** Across all five cells, $\Delta(\text{novel} - \text{LS})$ ranges from -0.001 to $+0.025$. The feature-stability penalty neither hurts nor materially helps at these qubit counts. This is the expected outcome here: the penalty was designed to combat Stage-2 feature drift in the BP regime, but at $n \leq 4$ the optimiser does not move the quantum parameters far enough to invalidate the Stage-1 readout in the first place. The novel variant is therefore a *safe drop-in* for vanilla LS and is expected to differentiate at $n \geq 6$, where the Stage-2 trajectory is much longer and feature drift becomes the binding constraint.

Take-away. The cleanest single-sentence summary of the whole study is: *two-stage LS is the right default mitigation for QLSTM at any depth $L \geq 2$; identity-block init is essentially equivalent to random init in the regime we can simulate; and our novel feature-regularised LS is a safe, theoretically motivated extension that we expect (but do not yet measure) to pull ahead at $n \geq 6$.*

10.4. Repository

All code, plots, and reports: <https://github.com/gautam2905/IQC-Project>

11. Future Direction

Characterising the trainability phase boundary. Trainability of a VQC is plausibly controlled not by qubit count alone but by a composite of expressibility, entanglement structure, and cost-function locality; crossing a sharp boundary in this composite space is what actually triggers a barren plateau. We would build a unified phase diagram by sweeping $n \in [2, 10]$, $L \in [1, 6]$, three entanglement topologies (chain / ring / all-to-all), and global vs. local measurements, recording $\text{Var}_\theta[\partial_\theta L]$ and final MAE under each mitigation in every cell. Fitting a single scaling law $\log \text{Var} \approx a - bn - cL - dE - e(1 - \ell)$ would subsume all four BP sources from Larocca et al. into one calibrated equation and pinpoint exactly where each mitigation begins to dominate. The phase diagram is a property of the underlying VQC ansatz, not the LSTM wrapper, so the contour transfers directly to QCNNs, quantum transformers, and VQE.

From a single mitigation to a learned mitigation policy. The four strategies studied here are isolated points in a much larger space of initialisation policies, and the right choice almost certainly depends on local geometry near θ_0 . We would frame mitigation selection as a contextual bandit whose context features are (i) gradient-variance estimate at θ_0 , (ii) condition number of the Stage-1 readout matrix $\Phi_0^\top \Phi_0$, and (iii) a finite-difference smoothness probe of the Stage-1 loss. Trained across a meta-distribution of (architecture, dataset) pairs with final test MAE as reward, this would yield a small decision tree that recommends the best mitigation in milliseconds for any new configuration — converting our static empirical benchmark into a deployable artefact and giving the first *prescriptive* answer to “which initialisation for this circuit, this data, this hardware?”

References

- [1] S. Chen, S. Yoo, and Y.-L. Fang. Quantum long short-term memory. *arXiv preprint arXiv:2009.01783*, 2020. <https://arxiv.org/abs/2009.01783>
- [2] M. Larocca et al. Barren plateaus in variational quantum computing: A review. *arXiv preprint arXiv:2405.00781*, 2024. <https://arxiv.org/abs/2405.00781>

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- [4] J. R. McClean, S. Boixo, V. N. Smelyanskiy, R. Babbush, and H. Neven. Barren plateaus in quantum neural network training landscapes. *Nature Communications*, 9(1):4812, 2018.
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